

Design a Secure Snowflake Environment for a Hypothetical Company

1. Introduction

This document presents the design and implementation of a secure Snowflake environment for a hypothetical company. The environment follows security best practices using Role-Based Access Control (RBAC), Network Policies, Multi-Factor Authentication (MFA), and Dynamic Data Masking to ensure data protection and controlled access.

2. Company Scenario

The company has three primary user roles:

Role	Description
Admin	Manages users, roles, warehouses, and databases
Analyst	Performs data analysis and has read access to data
Viewer	Accesses dashboards and reports only

3. User Creation

SQL Script

```
CREATE USER ADMIN_USER
PASSWORD = 'StrongPassword#123'
DEFAULT_ROLE = ADMIN_ROLE
MUST_CHANGE_PASSWORD = TRUE;
```

```
CREATE USER ANALYST_USER
PASSWORD = 'StrongPassword#123'
DEFAULT_ROLE = ANALYST_ROLE
MUST_CHANGE_PASSWORD = TRUE;
```

```
CREATE USER VIEWER_USER
PASSWORD = 'StrongPassword#123'
DEFAULT_ROLE = VIEWER_ROLE
MUST_CHANGE_PASSWORD = TRUE;
```

4. Role Creation and Assignment

SQL Script

```
CREATE ROLE ADMIN_ROLE;  
CREATE ROLE ANALYST_ROLE;  
CREATE ROLE VIEWER_ROLE;
```

```
GRANT ROLE ADMIN_ROLE TO USER ADMIN_USER;  
GRANT ROLE ANALYST_ROLE TO USER ANALYST_USER;  
GRANT ROLE VIEWER_ROLE TO USER VIEWER_USER;
```

5. RBAC Policy Design

5.1 Admin Role

Responsibilities

- User Management
- Role Management
- Warehouse Administration
- Database Administration

Privileges

```
GRANT CREATE USER ON ACCOUNT TO ROLE ADMIN_ROLE;  
GRANT CREATE ROLE ON ACCOUNT TO ROLE ADMIN_ROLE;  
GRANT CREATE DATABASE ON ACCOUNT TO ROLE ADMIN_ROLE;  
GRANT CREATE WAREHOUSE ON ACCOUNT TO ROLE ADMIN_ROLE;
```

5.2 Analyst Role

Responsibilities

- Data Analysis
- Query Execution
- Reporting

Privileges

```
GRANT USAGE ON DATABASE COMPANY_DB TO ROLE ANALYST_ROLE;  
GRANT SELECT ON ALL TABLES IN DATABASE COMPANY_DB TO ROLE ANALYST_ROLE;
```

5.3 Viewer Role

Responsibilities

- Dashboard Viewing
- Business Reporting

Privileges

```
GRANT USAGE ON SCHEMA COMPANY_DB.DASHBOARD TO ROLE VIEWER_ROLE;  
GRANT SELECT ON ALL TABLES IN SCHEMA COMPANY_DB.DASHBOARD TO ROLE  
VIEWER_ROLE;
```

6. Warehouse Access Configuration

```
GRANT ALL PRIVILEGES ON WAREHOUSE ADMIN_WH TO ROLE ADMIN_ROLE;
```

```
GRANT USAGE ON WAREHOUSE ANALYTICS_WH TO ROLE ANALYST_ROLE;
```

```
GRANT USAGE ON WAREHOUSE REPORTING_WH TO ROLE VIEWER_ROLE;
```

7. Network Policy Configuration

To restrict access to trusted corporate IP addresses, a network policy is implemented.

SQL Script

```
CREATE NETWORK POLICY CORP_POLICY  
ALLOWED_IP_LIST = (  
'203.0.113.10','203.0.113.11','203.0.113.12'  
);
```

```
ALTER ACCOUNT  
SET NETWORK_POLICY = CORP_POLICY;
```

Benefits

- Prevents unauthorized access
 - Allows connections only from trusted networks
 - Enhances account security
-

8. Multi-Factor Authentication (MFA)

Configuration Steps

1. Enable MFA in Snowflake.
2. Require MFA enrollment for all users.
3. Integrate with Identity Providers such as:
 - Microsoft Entra ID
 - Okta
 - Ping Identity
4. Disable MFA bypass where possible.

Example Configuration

```
ALTER ACCOUNT SET ALLOW_CLIENT_MFA_CACHING = FALSE;
```

Security Benefits

- Additional authentication layer
- Reduced risk of credential theft
- Improved compliance with security standards

9. Data Masking Implementation

Sensitive data such as Social Security Numbers (SSN) and Email Addresses should be protected using Dynamic Data Masking.

Create Masking Policy

```
CREATE OR REPLACE MASKING POLICY SSN_MASK_POLICY
AS (VAL STRING)
RETURNS STRING ->
CASE
WHEN CURRENT_ROLE() = 'ADMIN_ROLE'
THEN VAL
ELSE 'XXX-XX-XXXX'
END;
```

Apply Policy

```
ALTER TABLE CUSTOMER_DATA
MODIFY COLUMN SSN
SET MASKING POLICY SSN_MASK_POLICY;
```

10. Testing and Validation

Results (just now)

Table Chart Pivot

12 rows 74ms

	created_on	name	is_default	is_current	is_inherited	assigned_to_users	granted_to_roles	granted_roles	owner	comment	is_from_organization_user
		ACCOUNTADMIN 8.3% ADMIN 8.3% +10 more	N 91.7% Y 8.3%	N 91.7% Y 8.3%	N 66.7% Y 33.3%				ACCOUNTADMIN 50.0% ACCOUNTADMIN 50.0%	Account administrator can manage all aspects of the account 50.0% +5 more 8.3%	N
1	2026-05-25 02:56:01.979 -0700	ACCOUNTADMIN	Y	Y	N	1	0	2	ACCOUNTADMIN	Account administrator can manage all aspects of the account	N
2	2026-06-04 00:49:39.903 -0700	ADMIN	N	N	N	1	0	0	ACCOUNTADMIN		N
3	2026-06-04 02:57:36.642 -0700	ADMIN_ROLE	N	N	N	1	0	0	ACCOUNTADMIN		N
4	2026-06-04 00:49:40.895 -0700	ANALYST	N	N	N	2	0	0	ACCOUNTADMIN		N
5	2026-06-02 23:18:16.611 -0700	ANALYST_ROLE	N	N	N	1	0	0	ACCOUNTADMIN		N
6	2026-05-25 02:56:02.418 -0700	ORGADMIN	N	N	N	1	0	0		Organization administrator can manage organizations and a	N
7	2026-05-25 02:56:02.037 -0700	PUBLIC	N	N	Y	0	0	0		Public role is automatically available to every user in the acc	N
8	2026-05-25 02:56:02.014 -0700	SECURITYADMIN	N	N	Y	0	1	1		Security administrator can manage security aspects of the i	N
9	2026-05-25 02:56:02.024 -0700	SYSADMIN	N	N	Y	0	1	0		System administrator can create and manage databases an	N
10	2026-05-25 02:56:01.999 -0700	USERADMIN	N	N	Y	0	1	0		User administrator can create and manage users and roles	N
11	2026-06-04 00:48:42.255 -0700	VIEWER	N	N	N	0	0	0	ACCOUNTADMIN		N
12	2026-06-04 23:02:10.665 -0700	VIEWER_ROLE	N	N	N	1	0	0	ACCOUNTADMIN		N

Admin Access

Expected Result:

- Full visibility of sensitive data
- Administrative privileges available

Analyst Access

Expected Result:

- Read-only access to datasets
- Masked sensitive information

Viewer Access

Expected Result:

- Dashboard-only access
- Masked sensitive information
- No direct access to raw datasets

Table

Chart

Pivot

	# CUSTOMER_ID		A CUSTOMER_NAME	A SSN	A EMAIL
1		1	John Smith	XXX-XX-XXXX	john@email.com
2		2	Jane Doe	XXX-XX-XXXX	jane@email.com

11. Security Best Practices Implemented

- Role-Based Access Control (RBAC)
- Principle of Least Privilege
- Multi-Factor Authentication (MFA)
- Network Access Restrictions
- Dynamic Data Masking
- Warehouse Segregation
- Future Grants for New Objects

12. Conclusion

The designed Snowflake environment provides a secure, scalable, and compliant framework for managing organizational data. By implementing RBAC, network policies, MFA, and data masking, the company ensures that users have appropriate access while protecting sensitive information from unauthorized exposure.